

AHMED ALI TARIQ



Computer Science Graduate - NLP & Large Language Models

Multilingual NLP · Multimodal Learning · Generative Models · Explainable AI

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Research Focus

My research focuses on developing intelligent AI systems for language and multimodal understanding. I work on multilingual NLP, retrieval-augmented LLMs, and multimodal learning, with an emphasis on building accurate, explainable, and practical deep learning solutions.

01

Multilingual and cross-lingual NLP

Transformer-based models (XLM-R, mBERT) for classification and transfer across languages, with a focus on low-resource languages such as Urdu and Roman Urdu.

02

Multimodal and explainable AI

Multimodal models combining text and image features, with explainability methods (e.g., Grad-CAM for the visual branch) to understand how each modality influences predictions across languages.

03

Large Language Models & Generative AI

Exploring transformer-based large language models for language understanding, generation and instruction following, with applications in domain-specific and multilingual AI systems.

04

Retrieval-augmented LLMs

LLM systems that retrieve relevant information from external knowledge sources before generating responses, improving factual accuracy and enabling domain-specific question answering.

Education

B.S. Computer Science

HITEC University Taxila, Pakistan

Graduated · GPA 3.55 / 4.0 · Coursework: Artificial Intelligence, Data Structures, Digital Image Processing, Probability and Statistics, Numerical Computing

Research Projects

Explainable LLM-based Detoxification Framework for Low-Resource Languages

Ongoing Research · 2026 · Large Language Models

- Achieved 94.6% test accuracy, 91.6% fluency, and 92.9% semantic similarity, while reducing toxicity to 7%.
- Developed a toxicity classification component achieving 93.7% macro-f1, supporting reliable evaluation of detoxified outputs.

Bridging Modalities and Languages: An Explainable Cross-lingual Multimodal Deep Learning Framework for Hate Speech Detection

Manuscript Under Review · PeerJ Computer Science (Q1) · 2026

- End-to-end Multimodal PyTorch pipeline classifying Hate Speech across 3 languages on 11,000 instances - 88% test accuracy.
- Designed the full pipeline solo: pre-processing, balancing, training, evaluation, statistical significance tests and error analysis.

Multimodal Hate Speech Detection Interface using Advanced Deep Learning

Final-year thesis · Jun 2025 · mBERT + ViT

- Developed a multimodal hate speech detection model combining text and image features.
- Built an interactive interface for real-time multimodal analysis for hate speech predictions.

Research Trajectory

My near-term goal is to deepen my expertise in LLMs, multilingual NLP, and multimodal learning, with a focus on building explainable and reliable AI systems. Long-term, I aim to advance research in NLP and large language models, developing scalable, reproducible multimodal architectures that operate reliably across diverse languages and domains.

Technical Skills

PROGRAMMING Python · C++ · SQL	DEEP LEARNING PyTorch · scikit-learn · OpenCV · Hugging Face Transformers
NLP TECHNIQUES Cross-lingual transfer learning · multilingual transformers · LLM-assisted data generation and annotation	ML TECHNIQUES CNNs · Multimodal fusion · Attention mechanisms · Grad-CAM · data augmentation
DATA & ANALYSIS Pandas · NumPy · Experiment tracking · Statistical evaluation · Dataset curation and annotation	WRITING & REPRODUCIBILITY LaTeX · Markdown · Git/GitHub · Documented code · Peer-reviewed writing

Key Numbers

3.55 / 4.0 GPA - B.S. Computer Science, HITEC University	1st Author Paper under review, PeerJ Computer Science (Q1)	93.7% Macro-F1 Held-out test set, Urdu toxicity classification (XLM-R)	Urdu Dataset Parallel toxic/detoxified text dataset (low-resource NLP) 2,947 pairs
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Awards & Certifications

- Need Based Merit Scholarship - HITEC University · 2021-2025
- Machine Learning Specialization - Coursera certificate · 2025
- Mathematics for Machine Learning and Data Science Specialization - Coursera certificate · 2023

Languages

English <i>Medium of Instruction (MOI)</i> Certificate, HITEC University, (2025)	Urdu <i>Native</i>
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References

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